REPLY × LUISS · PROJECT 2 · 2026

Classical vs Multi-Agent Anomaly Detection

Two architectures, one detection problem — what changes, what stays the same.

TEAM
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NoiPA — Italian airport border alarms
567 routes · 54 engineered features · 13 months

The brief — what Reply asked

Not whether to do anomaly detection. With which software paradigm.

EXPLICIT ASKS

- Classical pipeline — sequential notebooks with IsolationForest, LOF, or Z-score on a historical baseline.
- Multi-agent system — 5 agents (Data, Baseline, Outlier, Risk, Report) with explicit shared-state communication.
- Architectural comparison — coherent detection, orchestration trade-offs, recommendation by use case.
- Deliverables — single end-to-end notebook, README, oral PPTX.

IMPLICIT (THE TRUE GAME)

- Numbers must converge, divergence means a bug. Convergence is the certificate of correctness.
- The architectural difference must show up elsewhere (operability, modularity, explainability, reuse, maintenance)
- The dataset is real but small (567×13), heavy temporal models like STL are dishonest.

The data — NoiPA, Italian airport borders

Two raw CSVs, joined on route × month → one analysis grain.

ALLARMI.csv

One row per check alarm: source (Interpol, SDI, NSIS), outcome, year-month.

TIPOLOGIA_VIAGGIATORE.csv

Per route-month: travellers entered, alarmed, investigated.

merge
route × year × month

ANALYSIS GRAIN

ROUTE = origin → destination

e.g. CMN → FCO (Casablanca → Rome Fiumicino)

567 unique routes

54 numeric features / row

BUSINESS QUESTION

Is this route doing something anomalous compared to the others?

What we delivered

One repo, two pipelines, shared core, runnable end-to-end.

main.ipynb

219 cells, 0 errors

Single end-to-end notebook: EDA, preprocessing, both pipelines, comparison, robustness.

classical_pipeline/

6 sequential steps

Orchestrator + step functions extracted from notebooks 01–06.

multiagent_pipeline/

5 agents + verifier

LangGraph: data, baseline, outlier, supervisor, risk, report.

shared/

single source of truth

preprocessing.py + FeatureBuilder used by both pipelines — no drift.

streamlit_app/

interactive demo

6 tabs, runtime perimeter filter, on-demand LLM narratives.

README + PPTX

audit-friendly

21 KB README with architecture map, key numbers, design rationale, file map.

Classical pipeline — 6 sequential steps

Linear orchestrator. Each step persists to disk. Audit-friendly. ~3 s end-to-end.

0 — Preprocess
merge 2 CSVs, fix types, ND codes

1 — Features
FeatureBuilder → 54 numeric / route

2 — Baseline
Robust MAD z-score + Tukey audit

3 — Detection
Ensemble: IF · LOF · Z · AE → p97/p90

4 — Risk Profile
5 business rules · 60% ML + 40% rules

5 — Evaluation
Silhouette · bootstrap · SHAP

Linear
Each step needs the previous.

Reproducible
random_state = 42 everywhere.

Fast
~3 s end-to-end, 567 routes.

Auditable
Every step persists CSV / JSON.

Multi-agent pipeline *(LangGraph)*

Same detection logic, restated as a graph of specialised agents over a shared state.

Supervisor (verifier)
optional · refit IF @ 3% on ALTA

Data
filter · features

Baseline
MAD z-score

Outlier
IF · LOF · Z · AE

Risk
5 business rules

Report
LLM narratives

SHARED STATE (LangGraph TypedDict) df_raw · df_features · df_baseline · df_anomalies · df_risk · report

Isolation

no agent imports another — communicate only via state

Modular

each agent re-runnable independently · failure-isolated

Operable

runtime perimeter filter · LLM only when needed · 1.6 s end-to-end

The 4 conditional edges — honest reading

2 truly route the common case · 2 are guard rails for degenerate input.

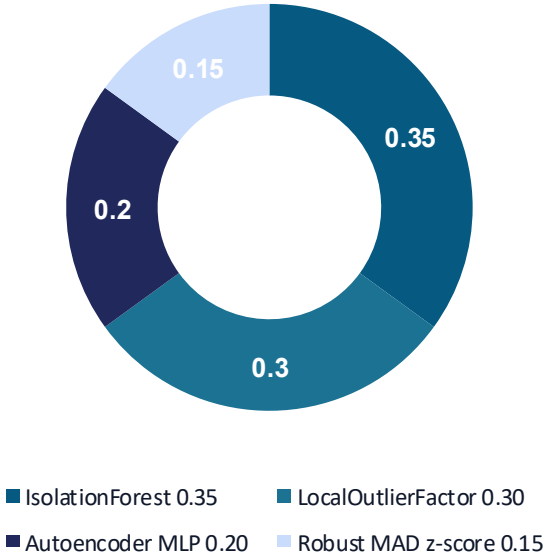
EDGE	DECISION	CHARACTER	FIRES ON 2024?
after_baseline	n_features < 5 OR std < 0.01 → END	Guard rail	No
after_outlier	n_alta ≥ 5 → Supervisor, else → Risk	Operative	Yes (17 ALTA)
after_supervisor	downgrade > 50% AND iter < 2 → cycle to Outlier, else Risk	Theoretical	No (29% rate)
after_risk	ALTA/MEDIA present AND run_report → Report, else END	Operative	Yes

Defending honestly is the point. A 2-route perimeter (filter by country = Algeria) must not crash — guard rails earn their keep there, not on the standard 2024 run.

The models — weighted ensemble + audit + surrogate

4 in the score · 1 audit-only · 1 SHAP surrogate.

ENSEMBLE WEIGHTS



0.35 · IF

IsolationForest

Tree-based, path-length isolation. Industry standard. n_estimators=200, contamination=0.10.

0.30 · LOF

LocalOutlierFactor

Density vs k-neighbours (k=20). Catches local anomalies near a cluster that IF misses.

0.20 · AE

Autoencoder (MLP 11→8→4→8→11)

Semi-supervised (normals from IF). Captures non-linear interactions IF/LOF/Z miss. Auto-disabled if <30 normals.

0.15 · Z

Robust MAD z-score

median ± 1.4826·MAD per feature, |z|.mean → score. Fallback to std when MAD=0 (sparse features).

5 business rules — the symbolic layer

br_score = mean. Combined with ML: 60% ML + 40% rules → confidence → final_risk.

RULE	CONDITION	BUSINESS MEANING
br_high_interpol	pct_interpol ≥ 0.30	Suspicious Interpol alarm rate.
br_high_rejection	tasso_respinti ≥ 0.25	Many rejections — active pattern.
br_low_closure	tot_allarmi_log > 3 AND tasso_chiusura < 0.10	Many alarms, almost none closed.
br_multi_source	pct_interpol > 0 AND pct_sdi > 0	Alarms from multiple sources — stronger signal.
br_high_alarm_rate	tasso_allarme_medio ≥ 0.50	Average alarm rate above operational threshold.

FINAL RISK `confidence = 0.6 · ensemble_score + 0.4 · br_score → { CRITICO · ALTO · MEDIO · BASSO }`

Six design choices we will defend

1

MAD over Tukey IQR for the ensemble

MAD is robust + composes into a scalar score. Tukey kept as audit (flag_<feature>) for analysts.

2

Ensemble of 4, not 1

Spec said "IF or LOF or Z". We added an Autoencoder for non-linear interactions. +15% runtime.

3

FeatureBuilder inside DataAgent (5, not 6)

Spec mandates 5 agents. Folding feature build into Data keeps topology honest.

4

No STL, no rolling-mean

13 months total, median route appears in 2. STL needs ≥12 obs. Replaced with linear trend slope per route.

5

Supervisor as verifier, not 6th agent

Refit IF @ contamination=3% on ALTA. Optional, fires only if ≥5 ALTA. Adds an alta_robusta tag.

6

LLM always-on for Report, lazy for Data

Pay LLM cost only where it adds value (narratives), not where deterministic code already works.

Results — detection on 2024 perimeter

567 routes, identical distribution under both architectures.

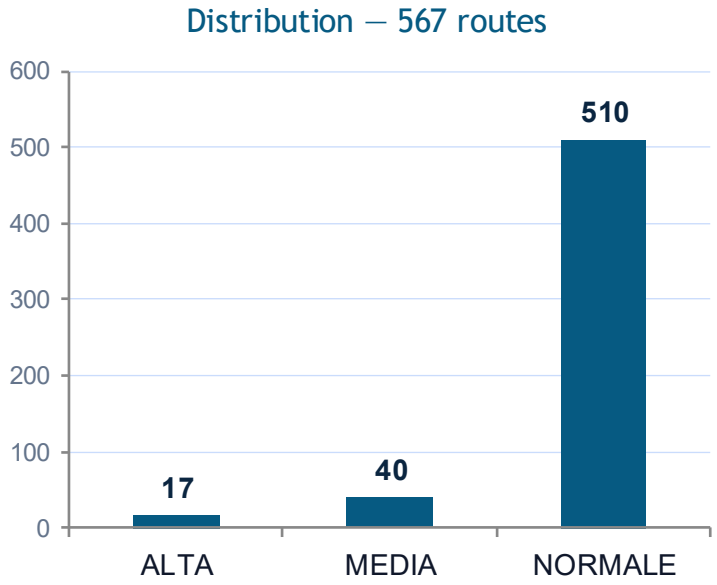
17	40	510
ALTA (high)	MEDIA (medium)	NORMALE (normal)

RUNTIME

Classical **~3.0 s**

Multi-agent **~1.6 s**

end-to-end · 567 routes · no LLM call



Convergence — 98.24% architectural agreement

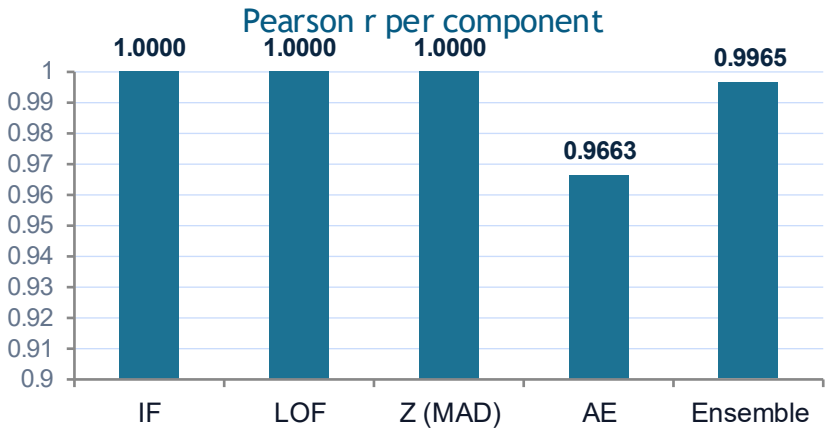
If the numbers diverged, one pipeline would be buggy. They don't. Cross-validation = free regression test.

LABEL AGREEMENT

98.24%

557 / 567 routes — identical label

residual 1.8% sits at the MEDIA / NORMALE boundary, entirely driven by AE training stochasticity.



Bootstrap CI

1000 resamples @ 80% on the agreement metric — implicit regression test.

Threshold sensitivity

Perturb the 5 BR thresholds by ±20% — measures decision fragility.

Trend slope

Linear slope per route on tasso_allarme_medio — pragmatic STL substitute.

RECOMMENDATION

Same numbers. Different strengths.

Convergence is the certificate of correctness. The decision is operational.

PICK CLASSICAL WHEN

- Single one-shot batch run, fixed perimeter.
- Auditor needs every intermediate CSV on disk.
- Team has no orchestrator stack and wants minimum dependencies.
- Lowest cognitive overhead for analysts reading the code.

PICK MULTI-AGENT WHEN

- Runtime perimeter changes (interactive Streamlit, what-ifs).
- Need granular re-run of one stage without restarting the chain.
- Failure isolation matters (LLM down ≠ pipeline down).
- Per-route narrative explanations are part of the deliverable.

Thank you · Questions?